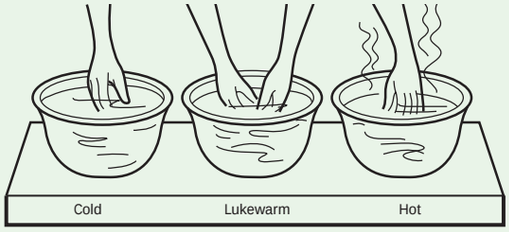
In short, Heat is the total kinetic energy of constituent particles of objects. **SI Unit of Heat is joule**. The unit calorie is also used.

1.3 Hot and cold objects

In our day-to-day life, we come across a number of objects. Some of them are hot and some of them are cold. How do we decide which object is hotter than the other?

We use the tip of our finger to find out whether the tea in a cup has enough heat to drink or whether milk has been cooled enough to set for making curds. We often determine heat by touching the objects. But is our sense of touch reliable?

Activity 1: Take three bowls. Pour very cold water in the first bowl. (you can also add ice cube for cooling). Place luke warm water in the second. Half fill the third with hot water (-not hot enough to burn!) Set them in a row on the table, with the lukewarm water in the center. Place your right hand in the cold water, and your left hand in the hot water. Keep them in for a few minutes. Then take them out, shake off the water and put both into the middle bowl. How do they feel?



Priya says,"My right hand tells me that the water in the bowl is hot and the left hand tells me that the same water is cold."

Write down in your own words what do you experience? Discuss in the class why this happens.

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When you placed your left hand in the hot tub, the heat from the bowl made the molecules on your hand vibrate faster. When you keep the same hot hand in the second bowl the vibrations transferred from your hand to make the particles in the water vibrate. Therefore you feel loss of heat and hence your hand feels cold.

In the same way, your right hand which was placed in cold water, feels hot when you insert it into the lukewarm water. Because it takes heat energy from lukewarm water.

So, the same lukewarm water gives your hands different feeling according to the temperature of your hand. **Measuring temperature by touching is not correct.**

Thermometers are used to measure temperature accurately and quantitatively.

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